2.3
1

a) $f(x) = \frac{1}{x}$ is not defined at $x = 0$.

b.) $f(x) = \sqrt{x}$ is not defined for $x < 0$.

c.) $f(x) = \pm\sqrt{(x^2+1)}$ does not map each x to a unique value, each x is mapped to two different values at once

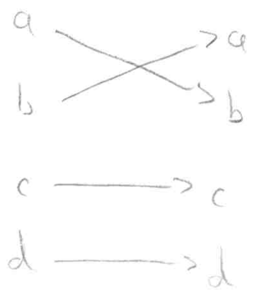
2.3
9

a) $\lceil \frac{3}{4} \rceil = 1$ b.) $\lfloor \frac{7}{8} \rfloor = 0$ c.) $\lceil -\frac{3}{4} \rceil = 0$ d.) $\lfloor -\frac{7}{8} \rfloor = -1$

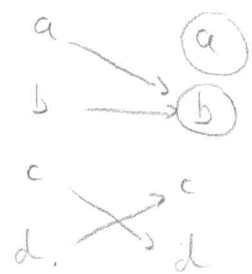
e.) $\lceil 3 \rceil = 3$ f.) $\lfloor -1 \rfloor = -1$ g.) $\lfloor \frac{1}{2} + \lceil \frac{3}{2} \rceil \rfloor = 2$ h.) $\lfloor \frac{1}{2} \cdot \lfloor \frac{5}{2} \rfloor \rfloor = 1$

2.3
10

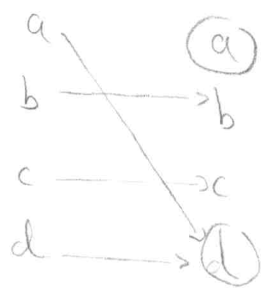
a.)



b.)



c.)



2.3
11

one-to-one ✓

onto ✓

not one-to-one X

not onto X

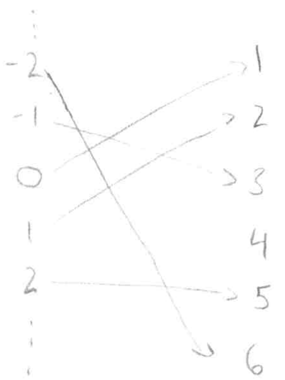
not one-to-one X

not onto X

2.3
21

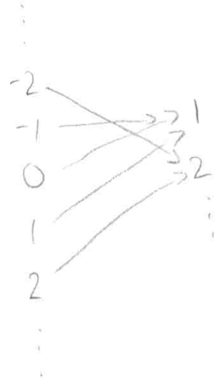
$f: \mathbb{Z} \to \mathbb{Z}^+$

a) one-to-one but not onto



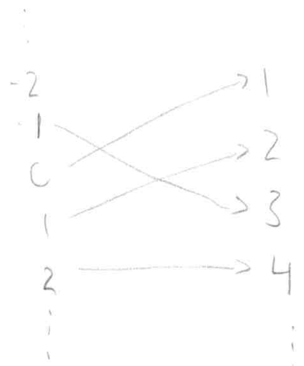
$$f(x) = \begin{cases} 3x - 1 & x > 0 \\ -3x & x < 0 \\ 1 & x = 0 \end{cases}$$

b) onto but not one-to-one



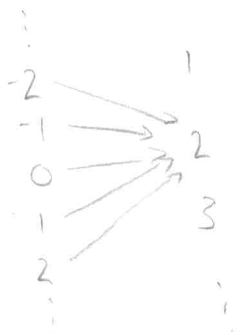
$$f(x) = \begin{cases} x & x > 0 \\ -x & x < 0 \\ 1 & x = 0 \end{cases}$$

c) one-to-one and onto



$$f(x) = \begin{cases} 2x & x > 0 \\ -2x + 1 & x \geq 0 \end{cases}$$

d) neither one-to-one nor onto



$$f(x) = 2$$

2.3 23

a) $f(x) = 2x+1$

$f: \mathbb{R} \to \mathbb{R}$

Suppose $f(x_1) = f(x_2)$

$2x_1 + 1 = 2x_2 + 1$

$2x_1 = 2x_2$

$x_1 = x_2$

so f is one-to-one.

Suppose $y \in \mathbb{R}$. $y = 2x+1$

$y - 1 = 2x$

$\frac{y-1}{2} = x$

So for any number $y \in \mathbb{R}$, f maps $\frac{y-1}{2}$ to that number, so f is onto

c) $f(x) = x^3$

Suppose $f(x_1) = f(x_2)$

$x_1^3 = x_2^3$

$x_1 = x_2$

So f is one-to-one.

Suppose $y \in \mathbb{R}$. $y = x^3$

$\sqrt[3]{y} = x$

So for any number $y \in \mathbb{R}$, f maps $\sqrt[3]{y}$ to that number, so f is onto.

b) not a bijection because $f(x) = x^2+1$ is not onto.

0 is in the codomain $\mathbb{R}$ but not in the range of f

d) not a bijection because $f(x) = \frac{x^2+1}{x^2+2}$ is not one-to-one

$f(1) = \frac{2}{3}$ and $f(-1) = \frac{2}{3}$.

but $1 \neq -1$

2.3 25 **Proof:** Suppose $f: \mathbb{R} \to \mathbb{R}$ and $f(x) > 0 \ \forall x \in \mathbb{R}$.

$\Rightarrow$) Suppose $f(x)$ is strictly decreasing.

This means if $x_1 < x_2$, then $f(x_1) > f(x_2)$.

Consider $g(x) = \frac{1}{f(x)}$. Suppose $x_1 < x_2$.

Since $f(x_1) > f(x_2)$, $g(x_1) = \frac{1}{f(x_1)} < \frac{1}{f(x_2)} = g(x_2)$.

So $g(x_1) < g(x_2)$, meaning $g(x)$ is strictly increasing.

$\Leftarrow$) Suppose $g(x)$ is strictly increasing.

This means if $x_1 < x_2$, $g(x_1) < g(x_2)$.

Consider $f(x) = \frac{1}{g(x)}$. Suppose $x_1 < x_2$

Since $g(x_1) < g(x_2)$, $f(x_1) = \frac{1}{g(x_1)} > \frac{1}{g(x_2)} = f(x_2)$

So $f(x_1) > f(x_2)$, so $f(x)$ is strictly decreasing

$\blacksquare$

2.3 26a

<u>Proof:</u> Suppose f is a strictly increasing function from $\mathbb{R}$ to $\mathbb{R}$.

This means if $x_1 < x_2$, then $f(x_1) < f(x_2)$.

We wish to show that f is one-to-one.

This means if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$.

So, suppose $x_1 \neq x_2$. This means that one of them must be larger than the other, so WLOG say $x_1 < x_2$.

Since f is strictly increasing, $f(x_1) < f(x_2)$.

$f(x_1) < f(x_2)$ means that $f(x_1) \neq f(x_2)$, so f is one-to-one.

2.3 36

<u>Proof:</u> Suppose f and $f \circ g$ are one-to-one.

For contradiction, suppose g is NOT one-to-one.

Then there exists x_1, x_2 such that $g(x_1) = g(x_2)$ but $x_1 \neq x_2$.

However, $f(g(x_1)) = f(g(x_2))$, and since $f \circ g$ is one-to-one, it must be the case that $x_1 = x_2$, a contradiction.

Thus, g must be one-to-one.

2.3 38

$f(x) = x^2 + 1$ $\quad$ $f(g(x)) = (x+2)^2 + 1 = \boxed{x^2 + 4x + 5}$

$g(x) = x + 2$ $\quad$ $g(f(x)) = (x^2 + 1) + 2 = \boxed{x^2 + 3}$

2.3 45

$g(x) = \lfloor x \rfloor$

a. $g^{-1}(\{0\}) = \boxed{[0, 1)}$

b. $g^{-1}(\{-1, 0, 1\}) = \boxed{[-1, 2)}$

c. $g^{-1}(\{x \mid 0 < x < 1\}) = \boxed{\emptyset}$ (floor must result in an integer)

2.3 53a

Proof: Suppose $x \in \mathbb{R}$, $n \in \mathbb{Z}$.

$\Rightarrow$): Suppose $x < n$. Since $\lfloor x \rfloor \leq x$, $\lfloor x \rfloor < n$

$\Leftarrow$) (Contrapositive) Suppose $x \geq n$.

Since $\lfloor x \rfloor$ is the largest integer less than or equal to x,

n cannot be larger than $\lfloor x \rfloor$.

Thus $\lfloor x \rfloor \geq n$.

$\blacksquare$

2.3 55

Proof: Suppose n is an integer

Case 1: Suppose n is even. Then $n = 2k$ for some $k \in \mathbb{Z}$.

Then $\frac{n}{2} = \frac{2k}{2} = k$, $\lfloor \frac{n}{2} \rfloor = \lfloor k \rfloor = k$.

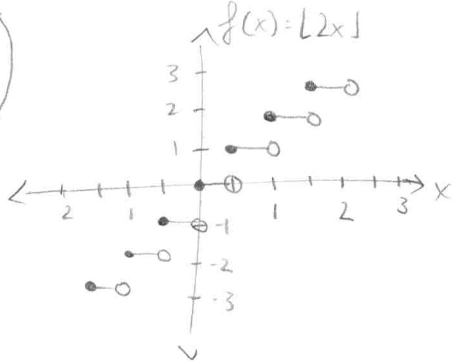
Thus $\frac{n}{2} = \lfloor \frac{n}{2} \rfloor$.

Case 2: Suppose n is odd. Then $n = 2k+1$ for some $k \in \mathbb{Z}$.

Then $\frac{n-1}{2} = \frac{2k+1-1}{2} = \frac{2k}{2} = k$ and $\lfloor \frac{n}{2} \rfloor = \lfloor \frac{2k+1}{2} \rfloor = \lfloor k + \frac{1}{2} \rfloor = k$

Thus $\frac{n-1}{2} = \lfloor \frac{n}{2} \rfloor$.

2.3 65



2.3 71

$f(x) = x^3 + 1$

$y = x^3 + 1$

$x = y^3 + 1$

$x - 1 = y^3$

$y = \sqrt[3]{x-1}$

$\boxed{f(x) = \sqrt[3]{x-1}}$

2.3
78

Proof: Suppose $x \in \mathbb{R}$. x must be between two integers, so $x \in [n, n+1)$ for some $n \in \mathbb{Z}$.

Thus, either $x \in [n, n+\frac{1}{3})$, $x \in [n+\frac{1}{3}, n+\frac{2}{3})$, or $x \in [n+\frac{2}{3}, n+1)$.

Case 1: Suppose $x \in [n, n+\frac{1}{3})$.

Thus, $3x \in [3n, n+\frac{1}{3})$, so $\lfloor 3x \rfloor = 3n$.

x, $x+\frac{1}{3}$, $x+\frac{2}{3}$ are each less than $n+1$, so $\lfloor x \rfloor = \lfloor x+\frac{1}{3} \rfloor = \lfloor x+\frac{2}{3} \rfloor = n$.

Thus, $\lfloor 3x \rfloor = \lfloor x \rfloor + \lfloor x+\frac{1}{3} \rfloor + \lfloor x+\frac{2}{3} \rfloor = 3n$.

Case 2: Suppose $x \in [n+\frac{1}{3}, n+\frac{2}{3})$

Thus, $3x \in [3n+1, 3n+2)$, so $\lfloor 3x \rfloor = 3n+1$

x, $x+\frac{1}{3}$ are each less than $n+1$, so $\lfloor x \rfloor = \lfloor x+\frac{1}{3} \rfloor = n$

$x+\frac{2}{3}$ must be at least $n+1$, so $\lfloor x+\frac{2}{3} \rfloor = n+1$.

Thus, $\lfloor 3x \rfloor = \lfloor x \rfloor + \lfloor x+\frac{1}{3} \rfloor + \lfloor x+\frac{2}{3} \rfloor = 3n+1$

Case 3: Suppose $x \in [n+\frac{2}{3}, n+1)$

Thus $3x \in [3n+2, 3n+3)$, so $\lfloor 3x \rfloor = 3n+2$

x is less than $n+1$, so $\lfloor x \rfloor = n$.

$x+\frac{1}{3}$, $x+\frac{2}{3}$ must be at least $n+1$, so $\lfloor x+\frac{1}{3} \rfloor = \lfloor x+\frac{2}{3} \rfloor = n+1$

Thus $\lfloor 3x \rfloor = \lfloor x \rfloor + \lfloor x+\frac{1}{3} \rfloor + \lfloor x+\frac{2}{3} \rfloor = 3n+2$

2.3
79

a.) $f: \mathbb{Z} \to \mathbb{R}$ $f(n) = \frac{1}{n}$

domain: $\boxed{\mathbb{Z}}$ codomain $\boxed{\mathbb{R}}$ domain of definition: $\boxed{\mathbb{Z} - \{0\}}$

undefined: $\boxed{\{0\}}$ <u>not total</u>

---

b.) $f: \mathbb{Z} \to \mathbb{Z}$ $f(n) = \lceil \frac{n}{2} \rceil$

domain: $\boxed{\mathbb{Z}}$ codomain: $\boxed{\mathbb{Z}}$ domain of definition: $\boxed{\mathbb{Z}}$

undefined: $\boxed{\emptyset}$ <u>total</u>

---

c.) $f: \mathbb{Z} \times \mathbb{Z} \to \mathbb{Q}$ $f(m,n) = \frac{m}{n}$

domain: $\boxed{\mathbb{Z} \times \mathbb{Z}}$ codomain: $\boxed{\mathbb{Q}}$ domain of definition: $\boxed{\mathbb{Z} \times (\mathbb{Z} - \{0\})}$

undefined: $\boxed{\mathbb{Z} \times \{0\}}$ <u>not total</u>

---

d.) $f: \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ $f(m,n) = n \cdot m$

domain: $\boxed{\mathbb{Z} \times \mathbb{Z}}$ codomain: $\boxed{\mathbb{Z}}$ domain of discourse: $\boxed{\mathbb{Z} \times \mathbb{Z}}$

undefined: $\boxed{\emptyset}$ <u>total</u>

---

e.) $f: \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ $f(m,n) = m-n$ if $m > n$

domain: $\boxed{\mathbb{Z} \times \mathbb{Z}}$ codomain: $\boxed{\mathbb{Z}}$ domain of discourse: $\boxed{\{(m,n) \mid m > n\}}$

undefined: $\boxed{\{(m,n) \mid m \leq n\}}$ <u>not total</u>

2.4 1 $a_n = 2(-3)^n + 5^n$

a) $a_0 = 2(-3)^0 + 5^0 = 2 + 1 = \boxed{3}$

b) $a_1 = 2(-3)^1 + 5^1 = -6 + 5 = \boxed{-1}$

c) $a_4 = 2(-3)^4 + 5^4 = 162 + 625 = \boxed{787}$

d) $a_5 = 2(-3)^5 + 5^5 = -486 + 3125 = \boxed{2639}$

2.4 4 a) $(-2)^n$

$a_0 = (-2)^0 = 1$

$a_1 = (-2)^1 = -2$

$a_2 = (-2)^2 = 4$

$a_3 = (-2)^3 = -8$

b) 3

$a_0 = 3$

$a_1 = 3$

$a_2 = 3$

$a_3 = 3$

c) $7 + 4^n$

$a_0 = 7 + 4^0 = 8$

$a_1 = 7 + 4^1 = 11$

$a_2 = 7 + 4^2 = 23$

$a_3 = 7 + 4^3 = 71$

d) $2^n + (-2)^n$

$a_0 = 2^0 + (-2)^0 = 2$

$a_1 = 2^1 + (-2)^1 = 0$

$a_2 = 2^2 + (-2)^2 = 8$

$a_3 = 2^3 + (-2)^3 = 0$

2.4 6b

$a_1 = 1$

$a_2 = 1 + 2 = 3$

$a_3 = 3 + 3 = 6$

$a_4 = 6 + 4 = 10$

$a_5 = 10 + 5 = 15$

$a_6 = 15 + 6 = 21$

$a_7 = 21 + 7 = 28$

$a_8 = 28 + 8 = 36$

$a_9 = 36 + 9 = 45$

$a_{10} = 45 + 10 = 55$

2.4 9e

$a_n = a_{n-1} + a_{n-3}$ $\quad a_0 = \boxed{1}$ $\quad a_1 = \boxed{2}$ $\quad a_2 = \boxed{0}$

$a_3 = a_2 + a_0 = 0 + 1 = \boxed{1}$

$a_4 = a_3 + a_1 = 1 + 2 = \boxed{3}$

2.4 13

$a_n = 8a_{n-1} - 16a_{n-2}$

a.) $a_n = 0 \quad a_{n-1} = 0 \quad a_{n-2} = 0$

$a_n = 8 \cdot 0 - 16 \cdot 0 = 0$

$0 = 0$ ✓

b.) $a_n = 1 \quad a_{n-1} = 1 \quad a_{n-2} = 1$

$a_n = 8 \cdot 1 - 16 \cdot 1 = 1$

$-8 \neq 1$ X

d.) $a_n = 4^n \quad a_{n-1} = 4^{n-1} \quad a_{n-2} = 4^{n-2}$

$a_n = 8 \cdot 4^{n-1} - 16\, 4^{n-2} = 4^n$

$2 \cdot 4^1 4^{n-1} - 4^2 \cdot 4^{n-2} = 4^n$

$2 \cdot 4^{n-1+1} - 4^{n-2+2} = 4^n$

$2 \cdot 4^n - 4^n = 4^n$

$4^n = 4^n$ ✓

g.) $a_n = (-4)^n \quad a_{n-1} = (-4)^{n-1} \quad a_{n-2} = (-4)^{n-2}$

$a_n = 8(-4)^{n-1} - 16(-4)^{n-2} = (-4)^n$

$-2 \cdot (-4)^1 \cdot (-4)^{n-1} - (-4)^2 (-4)^{n-2} = (-4)^n$

$-2 \cdot (-4)^{n-1+1} - (-4)^{n-2+2} = (-4)^n$

$-2(-4)^n - (-4)^n = (-4)^n$

$-3(-4)^n \neq (-4)^n \quad \times$

2.4 15d $\quad a_n = a_{n-1} + 2a_{n-2} + 2n - 9$

$a_n = 7 \cdot 2^n - n + 2 \qquad a_{n-1} = 7 \cdot 2^{n-1} - (n-1) + 2 \qquad a_{n-2} = 7 \cdot 2^{n-2} - (n-2) + 2$

$= 7 \cdot 2^{n-1} - n + 3 \qquad = 7 \cdot 2^{n-2} - n + 4$

$a_n = (7 \cdot 2^{n-1} - n + 3) + 2(7 \cdot 2^{n-2} - n + 4) + 2n - 9 = 7 \cdot 2^n - n + 2$

$= 7 \cdot 2^{n-1} - n + 3 + 7 \cdot 2^{n-1} - 2n + 8 + 2n - 9 = 7 \cdot 2^n - n + 2$

$= 2(7 \cdot 2^{n-1}) + (-n - 2n + 2n) + (3 + 8 - 9) = 7 \cdot 2^n - n + 2$

$= 7 \cdot 2^n - n + 2 = 7 \cdot 2^n - n + 2 \quad \checkmark$

2.4 19 $\quad$ a.) $a_n = 3a_{n-1}$

b.) $a_0 = 100$

$a_1 = 300$

$a_2 = 900$

$a_3 = 2700$

$a_4 = 8100$

$a_5 = 24300$

$a_6 = 72900$

$a_7 = 218700$

$a_8 = 656100$

$a_9 = 1968300$

$\boxed{a_{10} = 5904900}$

2.4
22

a) $a_n = 1.05 a_{n-1} + 1{,}000 \qquad a_0 = 50{,}000$

b) $a_1 = 53{,}500 \qquad a_5 = 69{,}339.71$

$a_2 = 57{,}175 \qquad a_6 = 73{,}806.69$

$a_3 = 61{,}033.75 \qquad a_7 = 78{,}497.03$

$a_4 = 65{,}085.44 \qquad \boxed{a_8 = 83{,}421.88}$ dollars

c)
$$a_n = 1.05\, a_{n-1} + 1000$$
$$= 1.05(1.05\, a_{n-2} + 1000) + 1000$$
$$= 1.05^2 a_{n-2} + 1.05 \cdot 1000 + 1000$$
$$= 1.05^2(1.05\, a_{n-3} + 1000) + 1.05 \cdot 1000 + 1000$$
$$= 1.05^3 a_{n-3} + 1.05^2\, 1000 + 1.05^1\, 1000 + 1.05^0\, 1000$$

same number

goes down by 1 each time

$$= 1.05^n \underbrace{a_{n-n}}_{= a_0} + 1.05^{n-1} \cdot 1000 + \cdots + 1.05^1 \cdot 1000 + 1.05^0 \cdot 1000$$
$$= 1.05^n\, 50{,}000 + \sum_{i=0}^{n-1} 1.05^i \cdot 1000$$
$$= 1.05^n \cdot 50{,}000 + \frac{1000 \cdot 1.05^{(n-1)+1} - 1000}{1.05 - 1}$$
$$= 1.05^n\, 50{,}000 + 1000\, \frac{1.05^n - 1}{0.05} = 1.05^n\, 50{,}000 + 20000 \cdot 1.05^n - 20000$$
$$= \boxed{70{,}000 \cdot 1.05^n - 20{,}000}$$

2.4 29c

$$\sum_{i=1}^{10} 3 = 10 \cdot 3 = \boxed{30}$$

2.4 30a

$\sum_{j \in S} j$ where $S = \{1, 3, 5, 7\}$

$$\sum = 1+3+5+7 = \boxed{16}$$

2.4 31c

$$\sum_{j=2}^{8} (-3)^j = \sum_{j=0}^{8} (-3)^j - (-3)^0 - (-3)^1$$

$$= \frac{(-3)^{8+1} - 1}{-3-1} - 1 + 3$$

$$= 4921 - 1 + 3 = \boxed{4923}$$

2.4 32c

$$\sum_{j=0}^{8} (2 \cdot 3^j + 3 \cdot 2^j)$$

$$= \sum_{j=0}^{8} (2 \cdot 3^j) + \sum_{j=0}^{8} (3 \cdot 2^j)$$

$$= \frac{2 \cdot 3^{8+1} - 2}{3-1} + \frac{3 \cdot 2^{8+1} - 3}{2-1}$$

$$= 19682 + 1533 = \boxed{21215}$$